Fe-C Problem 7 | Problems T4_PhaseDiagrams II.pdf #7 | Fe-C · hypereutectic cast iron

Hypereutectic Cast Iron from its Cementite Fraction

Original statement · Problems T4_PhaseDiagrams II.pdf, p. 2

7.- Consider an alloy of Fe-C, with content of C > 4.3%C at (727-ΔT)ºC. In these conditions, χ_{Fe_3C} = 0.73. Indicate the mass fraction of the phases at (1147+ΔT)ºC and (1147-ΔT)ºC.

Official Solution: T1: w_L = 0.75, w_{Fe_3C} = 0.25; T2: w_γ = 0.39, w_{Fe_3C} = 0.61, $w_{(Fe_3C)p}$ = 0.25, w_E = 0.75, $w_{\gamma E}$ = 0.39, $w_{(Fe_3C)E}$ = 0.36

PHASE 1 · HYPOTHESES & PARAMETERS

1. Equilibrium, Fe-Fe3C diagram [Session 9 Slide 4].
2. $C_0 > 4.3$: a **hypereutectic cast iron**; the primary phase is cementite, which forms in L + Fe3C above 1147 °C.
3. Unknown first: C_0 from the cementite fraction just below 727 °C.

PHASE 2 · FRAMES & STRATEGY

Tie lines: 727 − ΔT (α 0.022 / Fe3C 6.67), 1147 + ΔT (L 4.30 / Fe3C 6.67), 1147 − ΔT (γ 2.14 / Fe3C 6.67). Lever rule each time [Session 7 Slides 16, 32].

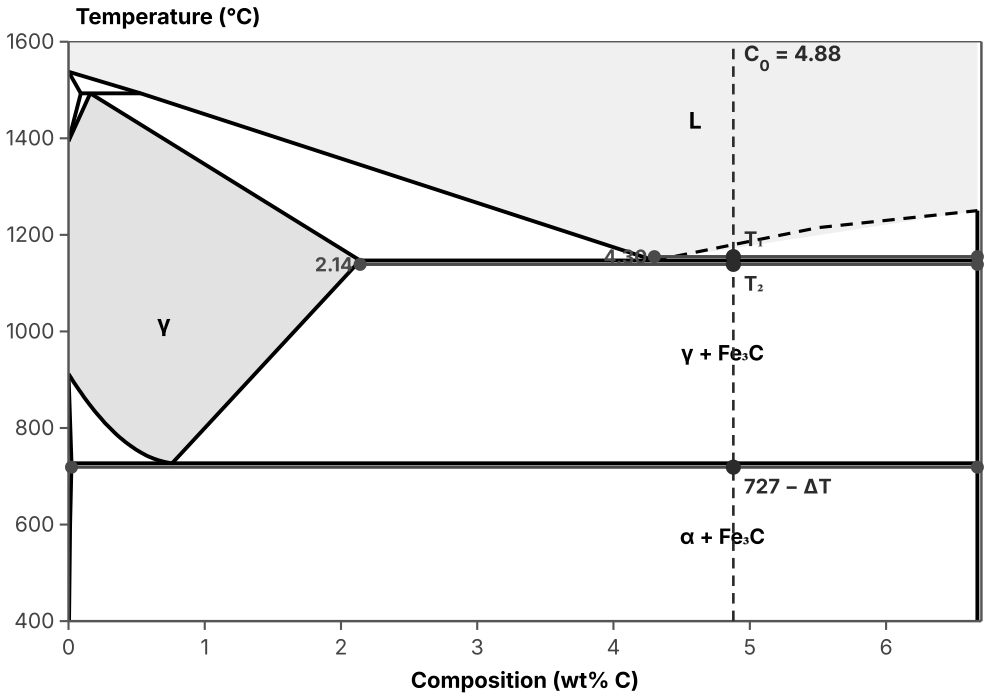


Figure 35. The alloy with 73 % Fe_3C just below 727 °C lies at about 4.88 % C: L + Fe_3C above 1147 °C, γ + Fe_3C below it.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Composition

$$C_0 = C_\alpha + W_{\mathrm{Fe_3C}}\,(C_{\mathrm{Fe_3C}} - C_\alpha) = 0.022 + 0.73 \times 6.648 = 0.022 + 4.853 = 4.875\ \%\,\mathrm{C}$$

2. T_1 **= 1147 + ΔT (length** $6.67 - 4.30 = 2.37$**)**

$$W_L = \frac{6.67 - 4.875}{2.37} = \frac{1.795}{2.37} = 0.757 \qquad W_{\mathrm{Fe_3C}} = \frac{4.875 - 4.30}{2.37} = \frac{0.575}{2.37} = 0.243$$

3. T_2 **= 1147 – ΔT (length** $6.67 - 2.14 = 4.53$**)**

$$W_\gamma = \frac{6.67 - 4.875}{4.53} = \frac{1.795}{4.53} = 0.396 \qquad W_{\mathrm{Fe_3C}} = \frac{4.875 - 2.14}{4.53} = \frac{2.735}{4.53} = 0.604$$

Microconstituents at T_2: primary cementite = 0.243; ledeburite = 0.757, made of γ 0.396 and Fe3C $0.604 - 0.243 = 0.361$.

PHASE 4 · CHECKS & INTERPRETATION

- Ledeburite make-up: $0.396 / 0.757 = 0.523$ γ (Worked example 14).
- Mass balance at T_1: $0.757 \times 4.30 + 0.243 \times 6.67 = 3.26 + 1.62 = 4.875$ % C .

Check against the official solution

Agrees with the key within rounding: 0.757/0.243 (key 0.75/0.25), 0.396/0.604 (0.39/0.61), primary Fe3C 0.243 (0.25), eutectic γ 0.396 (0.39) and eutectic Fe3C 0.361 (0.36). The key does not give C_0; it is 4.875 % C (4.89 % with 6.7 % C for cementite).